

* args vs ** kwargs

Base case
Condition

Tuple

Key-value Pairs

Recursive Portion Recursion → When a function call itself

$0!$ or $1! = 1$

$n \geq 0$

$5! \Rightarrow 5 * 4 * 3 * 2 * 1 = 120$

recursion

Loops

Every recursive code has its iterative solution as well.

```
def factorial(num):
```

```
    if num == 0 or num == 1:
```

```
        return 1
```

```
    else:
```

```
        return num * factorial(num-1)
```

Print(factorial(5))

120

Stack → LIFO

Last In First Out

Stack
empty

factorial(5) ¹²⁰

5 * factorial(4) ²⁴

4 * factorial(3) ⁶

3 * factorial(2) ²

2 * factorial(1) ¹

Stack fill
with the
sequence of
function calls

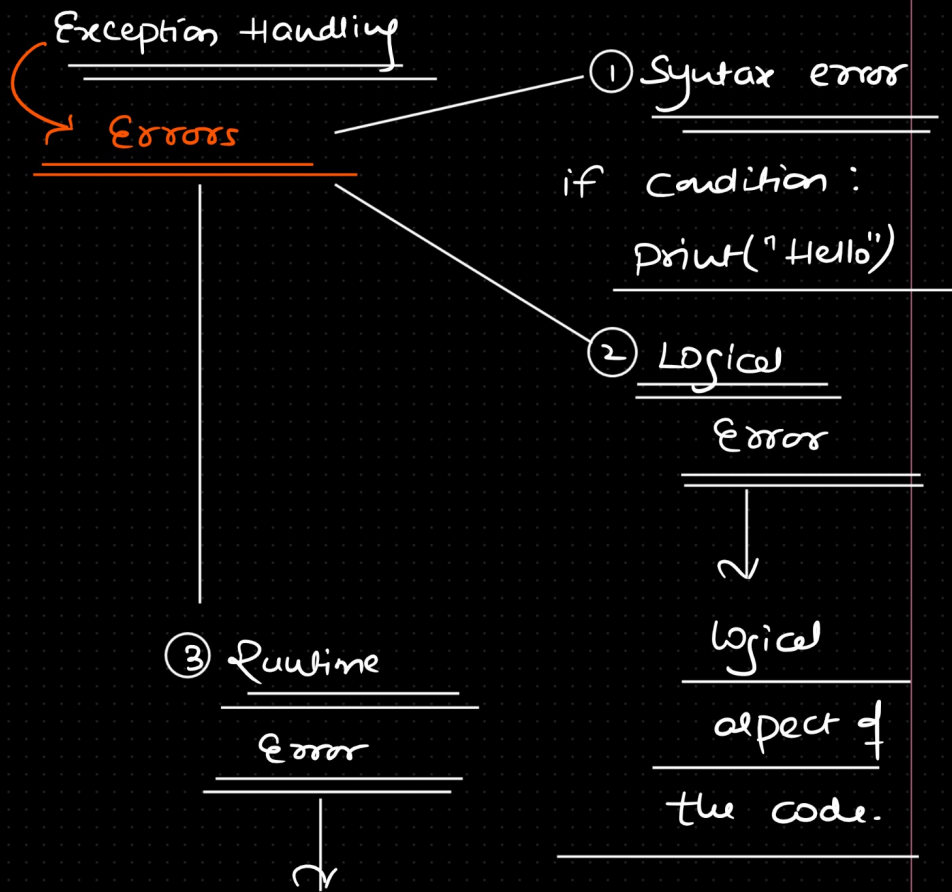
Key, value
0 1

item item[1]

lambda arg : expression

fact(1)	1
fact(2)	2
fact(3)	6
fact(4)	24
fact(5)	120

End → Stack empty



- ① try
- ② except
- ③ else
- ④ finally

SyntaxError: Occurs when the Python interpreter encounters an error in the syntax of the code, preventing it from executing. This can include issues like incorrect indentation, missing parentheses, or invalid keywords.

IndentationError: A specific type of SyntaxError that occurs when the indentation of code blocks is not consistent. Python uses indentation to define blocks of code, so incorrect or inconsistent indentation can lead to errors.

TypeError: Occurs when an operation or function is applied to an object of inappropriate type. For example, performing arithmetic on non-numeric data types or passing arguments of the wrong type to a function.

NameError: Occurs when a variable or function name used in the code is not defined or cannot be found in the current scope.

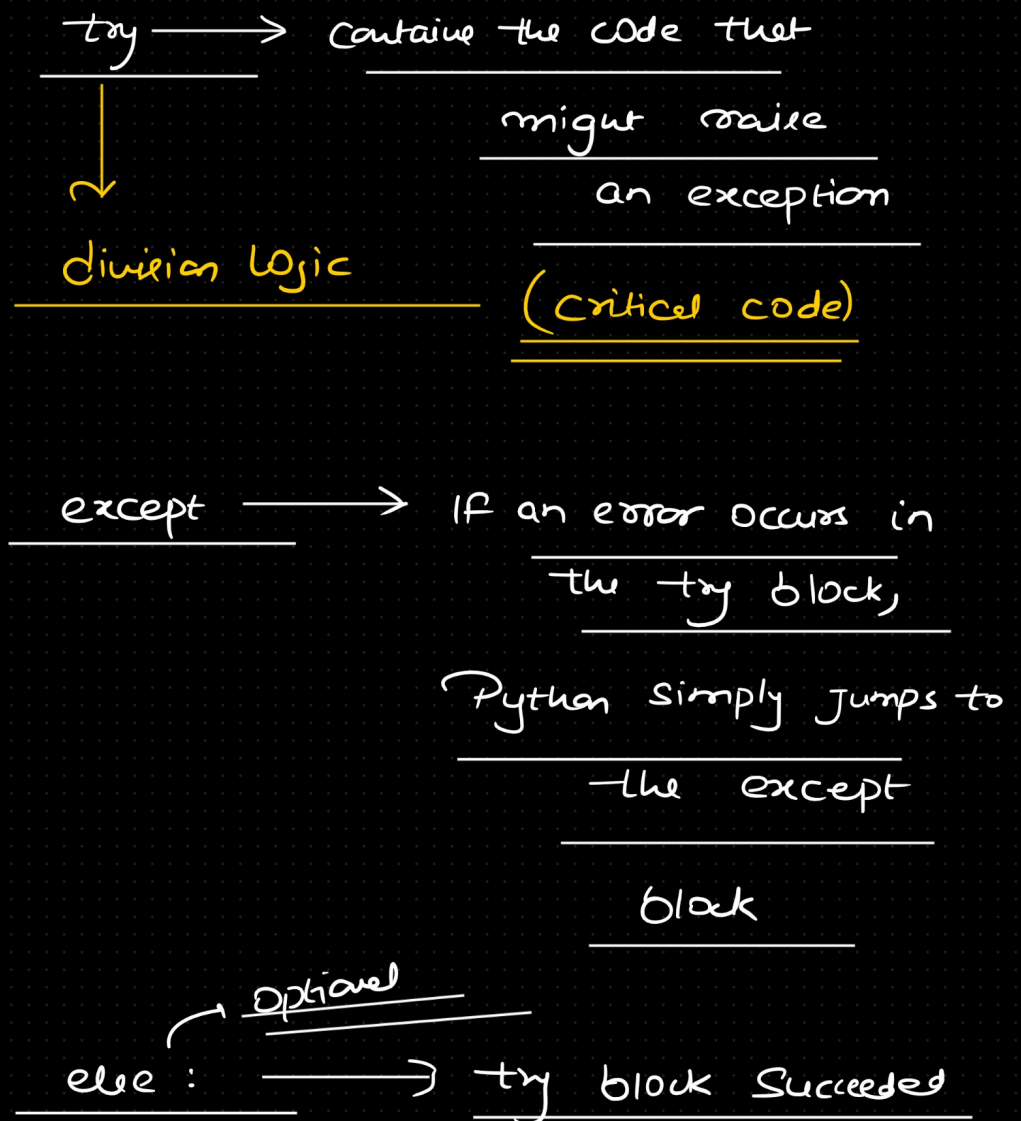
ValueError: Occurs when a function receives an argument of the right type but inappropriate value. For example, passing an invalid argument to a function that expects a certain range of values.

IndexError: Occurs when trying to access an element from a list, tuple, or string using an index that is out of its range. It indicates that the index provided does not exist in the sequence.

KeyError: Occurs when trying to access a dictionary with a key that does not exist in the dictionary. It indicates that the specified key is not found in the dictionary.

AttributeError: Occurs when trying to access an attribute or method that does not exist for a given object. It indicates that the object does not have the specified attribute or method.

ZeroDivisionError: Occurs when attempting to divide a number by zero, which is mathematically undefined. It indicates that the denominator in a division operation is zero.



try :

Print("Database start")

except :

finally :

Code → to close the
database